DC Charging (CCS2)-I

Introduction

As electric vehicles (EVs) become increasingly prevalent, the efficiency and safety of their charging infrastructure are paramount. While AC charging systems provide a convenient way to replenish EV batteries, they often involve power conversion within the vehicle, limiting charging speeds and requiring less complex communication. However, for rapid energy transfer, DC charging systems are essential, directly supplying power to the battery and necessitating a far more sophisticated communication protocol between the Electric Vehicle Supply Equipment (EVSE) and the EV.

This lecture delves into the intricacies of DC charging, specifically focusing on the Combined Charging System Type 2 (CCS2). Understanding CCS2 is crucial for anyone involved in EV technology, from engineers designing charging stations to users seeking to optimize their charging experience. It highlights the advanced communication mechanisms, such as Power Line Communication (PLC), and the multi-stage charging sequence that ensures safe, efficient, and high-power DC energy transfer, setting the foundation for the next generation of EV infrastructure.

Core Concepts

Term	Definition	Significance
`CCS2`	Combined Charging System Type 2, an EV charging standard that integrates both AC Type 2 and DC charging capabilities into a single vehicle inlet and charger plug.	Enables both AC and high-power DC charging using a unified physical interface, simplifying infrastructure and user experience.
`Power Line Communication (PLC)`	A technology that superimposes high-frequency data signals onto existing power lines (in this case, the Control Pilot line) to enable data transmission.	Facilitates high-level, bidirectional communication between the EV and EVSE, crucial for managing complex DC charging parameters.

Term	Definition	Significance
`PWM (Pulse Width Modulation)`	A modulation technique where the duty cycle of a square wave signal is varied to encode information. In EV charging, it's used on the Control Pilot line.	Provides a low-level communication channel, indicating basic states and maximum current limits, and serves as the carrier for PLC signals in DC charging.
`OSI Seven-Layer Model`	Open Systems Interconnection reference model, a conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers.	Ensures interoperability and robust communication protocols for high-level data exchange between EV and EVSE, similar to internet communication.
`HomePlug Green PHY`	A specific standard for Power Line Communication (PLC) that defines the physical interface for data transmission in EV charging applications.	Guarantees reliable and efficient data transfer over the CP line, forming the backbone of high-level communication in CCS2 DC charging.
`SLAC (Signal Level Attenuation Characterization)`	A procedure performed during the initial connection phase to characterize the signal attenuation between the EV and EVSE's PLC modems.	Ensures a stable and reliable logical connection between the EV and EVSE's communication systems before high-power charging commences.
`OFDM (Orthogonal Frequency Division Multiplexing)`	A digital modulation method that splits a high-speed data stream into several slower, parallel data streams, each transmitted on a different frequency subcarrier.	Enhances the robustness and efficiency of PLC by allowing data transmission across a wide frequency band with multiple subcarriers,

Term	Definition	Significance
		even in noisy environments.
`CP Line`	The Control Pilot line, a dedicated communication wire within the charging cable that carries PWM signals and, for DC charging, superimposed PLC signals.	The primary channel for both low-level and high-level communication, conveying critical charging status and control information.
`PP Pin`	The Proximity Pilot pin, a safety pin within the charging connector that detects whether the plug is correctly inserted and locked.	Ensures physical connection integrity and triggers vehicle immobilization, preventing accidental disconnections during charging.

Detailed Analysis

1. Recap of AC Type 2 Charging Communication

Before diving into DC charging, it's beneficial to recall the AC Type 2 charging communication. This system primarily uses a 1 kHz Pulse Width Modulation (PWM) signal on the Control Pilot (CP) line for low-level communication. The EVSE (Electric Vehicle Supply Equipment) changes the duty ratio of this PWM signal to indicate the maximum current that can be drawn, while the EV changes the voltage level of the PWM to signal its readiness.

The charging sequence for AC Type 2 involves several states:
* **State A**: EV not connected, CP voltage at +12V.
* **State B1**: EV connected, PWM deactivated, CP voltage at +9V. EVSE not ready, EV not ready for energy transfer. Resistance between CP and PE is 2740 ohms.
* **State B2**: EVSE ready, PWM activated (between +9V and -12V), EV not ready.
* **State C1**: EV ready, PWM deactivated (EVSE not ready), CP voltage at +6V. Resistance between CP and PE is 882 ohms.
* **State C2**: EV and EVSE ready, PWM activated (between +6V and -12V). Resistance between CP and PE is 882 ohms.
* **State D**: EV ready, requires ventilation, PWM activated (between +3V and -12V). Resistance between CP and PE is 246 ohms.

* **Fault Condition**: CP voltage goes to 0V or -12V, indicating a fault or EVSE unavailability.

The Proximity Pilot (PP) pin is used by the EV to detect if the correct connector is plugged in, based on the resistance (R6) value, which varies for different current ratings and AC/DC types. This detection allows the vehicle to recognize the plug's capabilities. The duty ratio of the PWM signal directly correlates to the maximum current the EVSE can provide, as per IEC 61851 standard:

Duty ratio (%)	Maximum current (A) to be drawn by EV
0 ≤ D ≤ 3	Not allowed
3 ≤ D ≤ 7	As indicated by digital communication (high-level communication) for DC charging
7 ≤ D ≤ 8	Not allowed
8 ≤ D ≤ 10	6
10 ≤ D ≤ 85	0.6 * D
85 ≤ D ≤ 96	(D - 64) * 2.5
96 ≤ D ≤ 97	80
97 ≤ D ≤ 100	Not allowed

In AC charging, power conversion occurs inside the vehicle, so the EVSE only needs to communicate its maximum current capability and readiness.

2. Transition to DC Charging (CCS2) and High-Level Communication

DC charging fundamentally differs from AC charging because the power conversion unit is located outside the vehicle, within the EVSE. This means DC power is directly fed into the EV's battery. Consequently, the EV's Battery Management System (BMS) must share extensive information with the EVSE's controller (Supply Equipment Communication Controller or SECC). This includes battery conditions like State of Charge (SOC), voltage, current, and temperature. Such detailed data exchange necessitates a dedicated, high-level communication channel, which for CCS2 is achieved through Power Line Communication (PLC) over the CP line.

3. CCS2 Connector and Communication Protocol

The CCS2 connector is a hybrid design. Its upper portion is identical to an AC Type 2 connector, allowing for AC charging. The lower portion adds two larger pins for DC+ and DC- connections. When a full CCS2 charger plug is inserted, it engages both AC Type 2 and DC pins. The communication for DC CCS2 charging still utilizes the CP and Protective Earth (PE) lines, but with a crucial difference: the PWM duty ratio is maintained between 3% and 7%. This specific duty ratio signals to the EV that it is a DC CCS2 charger, prompting the EV to enable high-level communication.

The high-level communication involves superimposing high-frequency signals onto the 1 kHz PWM signal on the CP line. This is where PLC comes into play. The EVSE and EV both have PLC modems that communicate over the CP line. This communication is structured according to the seven-layer OSI model, similar to how computers communicate over a network, ensuring robust and standardized data exchange. Key standards governing this communication include ISO 15118 and DIN SPEC 70121, which define message types, parameters, and timing.

4. Power Line Communication (PLC) Details and HomePlug Green PHY

The physical interface for PLC in CCS2 follows the HomePlug Green PHY standard, which is a subset of the broader HomePlug AV standards. Before any high-level communication begins, a Signal Level Attenuation Characterization (SLAC) procedure is performed. This ensures that the PLC modems of the EV and EVSE are logically connected and that the signal attenuation is minimal, guaranteeing reliable communication.

Key properties of HomePlug Green PHY based PLC:
* **Modulation**: Uses Orthogonal Frequency Division Multiplexing (OFDM) amplitude-phase modulation. OFDM separates a fast broadband transmission into many slower, narrowband transmissions, making it robust against interference.
* **Frequency Band**: Data is transmitted over a wide frequency band of 2 to 28 MHz.
* **Subcarriers**: Out of 1155 total subcarriers, 917 are actively used for data transmission.
* **Subcarrier Interval**: Each subcarrier frequency signal is generated with a 24.414 kHz interval.
* **Subcarrier Modulation**: Individual subcarriers are modulated using Quadrature Phase Shift Keying (QPSK).
* **Data Rate**: The digital data transmission rate ranges from 4.5 to 10 Mbps.
* **Signal Amplitude**: The peak-to-peak PLC signal is relatively small, ranging from 0.3Vpp to 1.3Vpp, superimposed on the 1 kHz PWM signal (which typically swings between +12V and -12V). This small amplitude necessitates the SLAC procedure to ensure signal integrity.

5. DC CCS2 Charging Sequence

The DC CCS2 charging sequence is a multi-stage process designed for safety and efficient high-power transfer:

- **State A (Unmated)**: Initially, the charger plug is disconnected. The CP line voltage is +12V.
- **T0: Mating and B1 State**: The charger plug is inserted into the EV.
- The EV detects the plug via the PP pin (R6 resistance), immobilizes the vehicle, and enters the B1 state (CP voltage +9V, PWM deactivated).
- **T0-T1: Initialization (B2 State)**:
- The EVSE activates PWM with a 3-7% duty ratio, signaling DC charging. The CP line is now in the B2 state (+9V, PWM activated).
- High-level communication (PLC) begins, and handshaking occurs.
- The EV sends its maximum voltage and current limits to the EVSE.
- A safety check is performed: if the DC voltage exceeds 60V or incompatibility is detected, an immediate shutdown occurs.
- **T1-T2: Connector Lock and Compatibility Check**:
- The EV locks the vehicle connector into its inlet.
- The EVSE responds to the EV with its maximum voltage capabilities.
- The EVSE performs an internal insulation check (no voltage applied to the connector yet).
- If the EV and EVSE are incompatible (e.g., voltage limits don't match), the vehicle will not proceed to the "Ready" state, leading to a shutdown.
- **T2-T3: Cable Check Stage (C2/D State)**:
- The EV changes its CP state from B2 to C2 (or D if ventilation is required), indicating it is "Ready" for energy transfer.
- The EV requests a cable and insulation check from the EVSE.
- The EVSE continuously checks and reports the high voltage system's isolation state.
- **T3-T4: Cable Check Completion**:
- The EVSE confirms that the isolation resistance of the system is above 100 kΩ.
- Upon successful isolation check, the EVSE indicates its status as "Valid" and "Ready" with a Cable Check Response.
- **T5: Pre-charge Stage**:

- With logical connection established and isolation confirmed, pre-charging begins.
- Contactors in the DC+ and DC- lines of the EVSE close.
- The EV, acting as the master, sends a "Pre-Charge Request" to the EVSE. This request includes the desired DC current (initially limited to less than 2A to manage inrush current) and the target DC voltage for pre-charging.

The subsequent stages, including full charging and unmating, build upon this intricate sequence, ensuring a controlled and safe high-power DC charging experience.

Key Takeaways

- [ ] DC CCS2 charging requires high-level communication due to external power conversion and extensive battery data exchange.
- [ ] The CCS2 connector integrates both AC Type 2 and DC pins, allowing for versatile charging options.
- [ ] A 3-7% PWM duty ratio on the CP line signals DC charging, initiating high-level communication.
- [ ] Power Line Communication (PLC) superimposes high-frequency data signals onto the 1 kHz PWM on the CP line.
- [ ] The HomePlug Green PHY standard, along with SLAC, governs the physical layer of PLC for reliable data transmission.
- [ ] The DC CCS2 charging sequence involves multiple safety-critical stages, including mating, initialization, connector locking, cable checks, and pre-charging.
- [ ] The EV's BMS communicates vital battery parameters (SOC, voltage, current, temperature) to the EVSE during DC charging.
- [ ] Standards like ISO 15118 and DIN SPEC 70121 define the detailed communication protocols based on the OSI seven-layer model.

Conclusion

The detailed communication protocols and multi-stage sequence of DC CCS2 charging underscore the complexity and sophistication required for high-power energy transfer to electric vehicles. Unlike simpler AC charging, DC charging demands a constant, rich dialogue between the EV and EVSE, facilitated by technologies like Power Line Communication and standardized by frameworks such as the OSI model and HomePlug Green PHY. This intricate dance of signals and data ensures not only the efficiency of charging but, more critically, the safety of both the vehicle and its occupants.

As EV technology continues to advance, with higher battery capacities and faster charging demands, the robustness and intelligence of these communication systems will become even more vital. Understanding these foundational principles of DC CCS2 charging is essential for developing future charging solutions that are faster, safer, and seamlessly integrated into our evolving energy landscape. The remaining stages of the charging process, including the full charging phase and eventual disconnection, further elaborate on this sophisticated interplay, ensuring a comprehensive and secure charging experience.